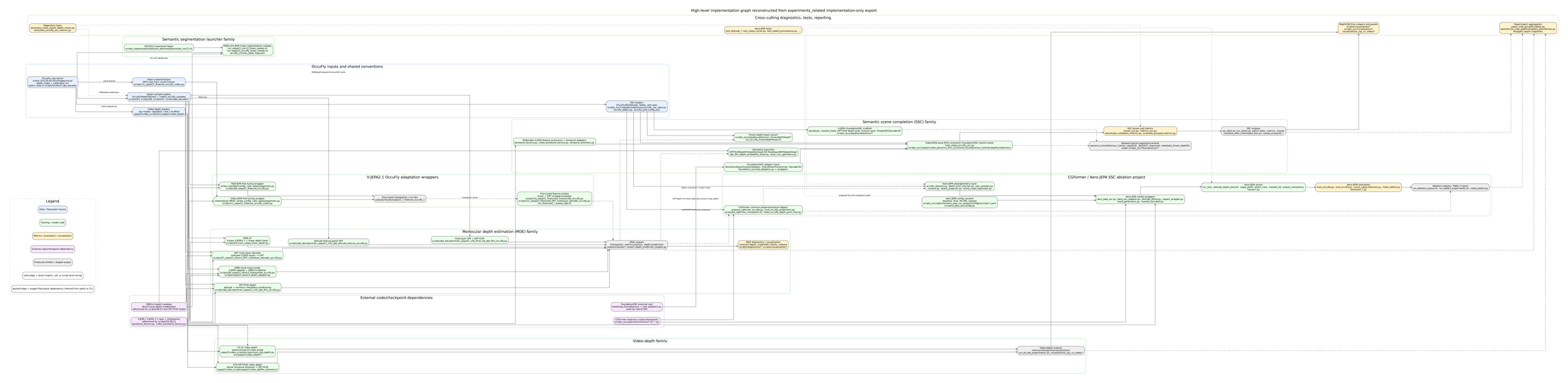

High-level implementation graph reconstructed from experiments_related implementation-only export

Cross-cutting diagnostics, tests, reporting
Repository tests
tests/test_mask_aware_depth_resize.py
tests/test_occufly_ssc_metrics.py
Aero-JEPA tests
test_altitude_*, test_vjepa_cache.py, test_table3_provenance.py, ...
Depth/SSC/live viewers and panels
scripts/visualization/*
scripts_ssc/vizualization/*
visualizations_rgv_vs_video/*
Paper/report aggregation
paper_mde_grouped_tables.py
best20rmse_mde_pipeline/update_best20rmse.py
forpaper_report snapshots
Semantic segmentation launcher family
VOC2012 download helper
scripts_segmentation/dataset_download/download_voc12.sh
DINOv3/V-JEPA linear segmentation sweeps
run_vjepa21_voc12_linear_sweep.sh
run_vjepa21_occufly_linear_sweep.sh
occufly_21class_label_map.json
OccuFly labels/map
OccuFly inputs and shared conventions
RGB/depth/poses/intrinsics/SSC grids
OccuFly raw layout
scene_XX/{30,40,50}/images/visual
depth_maps + calibration.txt
paths cited in scripts/03,06,07,dpt_decoder
visual frames
Video materialization
MP4 clips from visual frames
scripts/11_vjepa21_finetune_occufly_video.py
RGB/depth/calibration
Depth sample loaders
OccuFlyDepthDataset + collect_occufly_samples
scripts/03, scripts/06, scripts/07, scripts/dpt_decoder/*
RGB only
frame sequences
Video depth loaders
clip modes: repeated / real / shuffled
vjepa21video_scripts/src/vjepa21video_depth
SSC loaders
OccuFlySSCDataset, labels, grid spec
scripts_ssc/*/vjepafoundationssc/occufly_ssc_data.py
occufly_labels.py, occufly_grid_config.json
Semantic scene completion (SSC) family
V-JEPA FoundationSSC scaffold
backbone, context_head, DPT-FiLM depth prob, frustum pool, SimpleSSCDecoder3D
scripts_ssc/vjepafoundationssc/*
Frozen-depth-head variant
scripts_ssc/vjepafoundationssc_frozendepthhead/*
run_occufly_frozendepthhead.sh
RGB/video V-JEPA feature extractors + temporal adapters
backbone_factory.py, video_backbone_factory.py, temporal_attention.py
Video/RGB pose-MVS consistent FoundationSSC hybrid trainer
train_vjepa_occufly_ssc.py
scripts_ssc/vjepa21video_posemvs_film_consistent_foundationssc_hybrid/vjepafoundationssc
Geometry branches
DPTFiLMDepthProbabilityHead OR PoseAwareMVSDepthHead
dpt_film_depth_probability_head.py, pose_mvs_geometry.py
FoundationSSC adapter stack
GeometryAwareContextAdapter, HybridViewTransformer, Decoder3D
foundation_ssc/real_adapters.py + wrappers
SSC losses and metrics
losses_ssc.py, metrics_ssc.py
reevaluate_validation_metrics.py, evaluate_grouped_metrics.py
SSC outputs
ssc_best.pt, ssc_latest.pt, epoch ckpts, metrics, visuals
visualize_after_meaningful_test.py; sweep_overprob_*
Related hybrid snapshots/variants
posemvs_foundationssc_hybrid, vjepaF2D...BACKUP, improved, metadata_frozen_depthfix
under scripts_ssc/*foundationssc*
V-JEPA2.1 OccuFly adaptation wrappers
RGB JEPA fine-tuning wrapper
writes manifest/config; calls vjepa2/app/main.py
scripts/08_vjepa21_finetune_occufly.py
Video JEPA fine-tuning wrapper
materializes MP4s; writes config; calls vjepa2/app/main.py
scripts/11_vjepa21_finetune_occufly_video.py
Fine-tuned checkpoints / run dirs
outputs/results/vjepa21_*_finetune_occufly_*
checkpoint reuse
Fine-tuned feature probes
linear + DPT wrappers/sweeps
scripts/10_vjepa21_finetuned_linearprobe_occufly.py
scripts/11_vjepa21_finetuned_DPT_multilayer_decoder_occufly.py
run_finetuned_*_sweep_ddp.sh
CGFormer / Aero-JEPA SSC ablation project
Aero-JEPA data/geometry layer
occufly_dataset.py, depth_prior_teacher.py, pair_sampler.py
camera.py, latent_projector.py, world_voxel_alignment.py
depth checkpoint / frozen head
DPT-depth init when geometry_branch=dpt_depth
prepared OccuFly trees/prior paths
Aero-JEPA config variants
baseline, final, M1-M6, sweeps
scripts_ssc/cgformer/aero_jepa_ssc_project/configs/occufly/*.yaml
src/aero_jepa_ssc/config.py
V-JEPA/DPT-FiLM priors exported
CGFormer common preparation/eval helpers
prepare_cgformer_occufly.py, eval_occufly_predictions.py
evaluate_cgformer_checkpoint.sh, make_occufly_depth_prior_tree.py
Aero-JEPA model wrappers
aero_jepa_ssc.py, base_ssc_adapter.py, altitude_lifting.py, vjepa2_wrapper.py
mask_generators.py, masked_3d_head.py
Aero-JEPA losses
ssc_loss, altitude_depth_teacher, vjepa_distill, latent_view, masked_3d, output_consistency
losses/*.py
Aero-JEPA tools/tests
train_occufly.py, eval_occufly.py, cache_vjepa_features.py, make_table3.py
tests/test_*.py
Ablation outputs / Table 3 inputs
run_ablation_queue.sh, run_table3_experiments.sh, make_table3.py
Legend
Data / filesystem layout
Training / model code
Metrics / evaluation / visualization
External repo/checkpoint dependency
Produced artifact / staged output
solid edge = direct import, call, or script-level wiring
dashed edge = staged file/output dependency inferred from paths or CLI
Monocular depth estimation (MDE) family
MDE-LP
frozen V-JEPA2.1 + linear depth head
scripts/03_train_vjepa_linear_depth.py
Altitude feature-plane DPT
scripts/dpt_decoder/train_vjepa21_vitG_dpt_altitude_feature_occufly.py
Final-layer SFP + DPT-FiLM
scripts/dpt_decoder/train_vjepa21_vitG_final_sfp_dpt_film_occufly.py
MDE outputs
checkpoints, metrics.json/csv, depth predictions
outputs/results/*; export_depth_prediction_targets.py
MDE diagnostics / visualization
constant depth, scale/shift checks, viewers
scripts/diagnostics/*, scripts/visualization/*
DPT multi-layer decoder
selected V-JEPA layers -> DPT
scripts/07_vjepa21_dinov3_DPT_multilayer_decoder_occufly.py
DINO-style linear probe
V-JEPA adapter + DINOv3 depther
scripts/06_vjepa21_dinov3_linearprobe_occufly.py
scripts/vjepa21_dinov3_depth_adapter.py
DPT-FiLM depth
altitude + intrinsics metadata conditioning
scripts/dpt_decoder/train_vjepa21_vitG_dpt_film_occufly.py
External code/checkpoint dependencies
DINOv3 depth modules
dinov3.eval.depth.models/loss
referenced by scripts/06,07 and DPT-FiLM heads
V-JEPA / V-JEPA 2.1 repo + checkpoints
referenced by scripts/03,08,11
backbone_factory.py, video_backbone_factory.py
FoundationSSC external root
bootstrap_foundationssc + real_adapters.py
used by hybrid SSC
CGFormer external scripts/checkpoints
scripts_ssc/cgformer/common/*.sh, *.py
Video-depth family
ST-LP video depth
spatio-temporal linear probe
vjepa21video_scripts/scripts/train_stlp_depth.py
src/vjepa21video_depth/*
DTA DPT-FiLM video depth
dense temporal attention + DPT-FiLM
vjepa21video_scripts/vjepa21video_dptfilm_attention/*
Video-depth outputs
metrics/checkpoints/visualizations
run_all_stlp_experiments.sh; visualizations_rgv_vs_video/*